

Erdős Problem #36

EP-36	L1 - Tractable	Ouvert	Number Theory
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Énoncé original (anglais, reproduction mot pour mot)

Find the optimal constant $c > 0$ such that the following holds.
For all sufficiently large N , if $A \sqcup B = \{1, \dots, 2N\}$ is a partition into two equal parts, so that $|A| = |B| = N$, then there is some x such that the number of solutions to $a - b = x$ with $a \in A$ and $b \in B$ is at least cN .

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